ORION ASTRATHON

Problem Statement

Hyper-Local Aurora Forecasting & Astrophotographer Intelligence Platform

Domain	Duration	Difficulty
Space Weather & Geospatial	**48 Hours**	**Advanced / Stretch**

1. Background

Auroral phenomena — the Aurora Borealis and Aurora Australis — represent one of nature's most visually dramatic events. They arise from a complex electromagnetic interaction between the solar wind and Earth's magnetosphere, producing cascading light displays that can span entire hemispheres. Yet for the scientists, astrophotographers, and citizen observers who chase them, predicting exactly when and where an aurora will be visible remains a frustrating, imprecise endeavor.

1.1 The Science Behind the Aurora

The solar wind — a continuous plasma stream of electrons and protons ejected from the sun's corona at 300–800 km/s — carries an embedded Interplanetary Magnetic Field (IMF). The key driver of auroral activity is the Bz component of this field. When Bz turns negative (southward), magnetic reconnection occurs at the magnetopause, allowing solar wind particles to funnel along field lines toward the poles. These particles collide with atmospheric oxygen and nitrogen at altitudes of 100–300 km, exciting electrons that emit photons at characteristic wavelengths — the familiar greens (557.7 nm, atomic oxygen), reds (higher altitude oxygen), and purples (molecular nitrogen).

Auroral intensity is indexed using the Kp scale (0–9), a quasi-logarithmic global average computed from 13 magnetometer stations worldwide. Higher Kp values indicate southward auroral ovals extending to lower latitudes:

Kp Index	Storm Level	Approx. Visibility Latitude	Photography Outlook
0–2	Quiet	65°+ (Arctic only)	Dim; visible only at very high latitudes
3–4	Active	58°–65°	Formations develop; rural photographers activated
5	G1 Minor Storm	~56°	Northern US / Southern UK border region
6	G2 Moderate	~54°	Mid-latitude displays; bright and active
7	G3 Strong	~52°	Visible over Oregon, Central UK
8–9	G4–G5 Extreme	48°–50°	Overhead at mid-latitudes; rare, intense events

1.2 The Gap in Existing Tools

Current aurora forecasting tools — such as Hello Aurora, My Aurora Forecast, and NOAA's own SWPC dashboard — suffer from several well-documented limitations that create a clear opportunity for innovation:

- Over-reliance on the Kp index, which is a 3-hour average. It systematically misses brief but intense substorm events lasting 10–30 minutes, which represent the most photogenic aurora displays.
- No integration of photographer-specific constraints. Cloud cover, light pollution (Bortle scale), lunar phase, and local viewing geometry are treated as separate concerns, leaving users to manually cross-reference multiple apps.
- No hyper-local intelligence. Global oval maps lack the resolution to tell a user in Edinburgh whether they can see aurora from their backyard versus driving 40 km north.
- No logistical routing. No current tool suggests an optimal driving route to reach a clear-sky, dark-sky site within the aurora viewing zone.

2. Problem Statement

Design and build a real-time, hyper-local aurora forecasting and visualization platform that bridges the gap between raw space weather telemetry and on-the-ground visibility intelligence for astrophotographers, researchers, and enthusiasts.

Core Problem

Existing aurora forecasting tools provide global-scale indices and static oval maps, but fail to answer the question that matters most to an observer on the ground: "Will I be able to see and photograph the aurora from my location, right now, and what do I need to do to maximize that chance?"

2.1 What You Must Solve

Your solution must address the following interconnected challenges:

Real-Time Solar Wind Ingestion & Threshold Intelligence

Move beyond Kp. Ingest live solar wind velocity and IMF Bz data from NOAA's DSCOVR/ACE satellite feeds. Build threshold-based nowcasting logic that can fire alerts when Bz < –7 nT or solar wind speed exceeds 500 km/s — even if these conditions last only minutes and never register in the 3-hour Kp average.

OVATION-Prime Probability Visualization

Consume the NOAA OVATION auroral probability grid (360×181 resolution) and render it as an interactive, geospatially accurate overlay on a globe or map interface. This must not be a static image — the map must update automatically and reflect the current magnetic midnight sector as Earth rotates.

Photographer Visibility Scoring

Build a composite Visibility Score for any user location that weighs three key constraints: aurora probability (from OVATION), cloud cover (from meteorological APIs), and darkness (combining Bortle class light pollution data with real-time lunar illumination and the day/night terminator).

Stretch Goal — Logistical Routing

Calculate a GPS-navigable route from the user's current location to the nearest point that satisfies all three visibility criteria simultaneously: within the auroral viewing zone, below 30% cloud cover, and below Bortle Class 4. This is a stretch goal for teams that complete the core objectives.

2.2 Scope and Constraints

Your solution must function as a usable web application or mobile-responsive interface. It must ingest live data (not static datasets) and refresh automatically at appropriate cadences. The interface must be designed for real-world field use — legible in low-light conditions, intuitive under time pressure, and mindful of limited cellular connectivity in rural areas.

Scope is intentionally ambitious. Teams are not expected to complete every component — prioritize depth and quality in your core modules over a thin implementation of all features.

3. Expected Deliverables

All teams must submit the following by the end of the 48-hour window:

3.1 Core Deliverables (Required)

#	Deliverable	Description
D1	Live Data Pipeline	A backend service that continuously polls NOAA SWPC endpoints and processes solar wind and OVATION data. Must handle schema changes, failover between DSCOVR and ACE, and flag data gaps.
D2	Interactive Aurora Map	A web-based globe or 2D map rendering the OVATION probability grid as a dynamic, color-ramped overlay. Must include a day/night terminator layer and update automatically.
D3	Visibility Score Engine	An algorithm that computes a per-location visibility score (0–100) combining aurora probability, cloud cover, and darkness. Must be queryable for any lat/lon in real time.
D4	Alert System	A notification mechanism (push, in-app, or email) that fires when visibility score exceeds a user-configurable threshold at a saved location.
D5	Working Demo + Codebase	A deployed or locally runnable demo accessible during judging. Full source code in a public or shared repository with a clear README.

3.2 Stretch Deliverables (Bonus)

- GPS Routing: Geospatial routing to the nearest high-visibility location meeting all three criteria (aurora probability > 50%, cloud cover < 30%, Bortle < 4).
- Substorm Early Warning: Real-time monitoring of the negative Bz deflection rate to fire a 10-minute substorm precursor alert.
- Community Sighting Layer: Allow users to submit geo-tagged sighting reports (with photos) that appear as real-time validation pins on the map.
- Photography Settings Advisor: Based on the current Kp and aurora brightness estimate, recommend ISO, aperture, and shutter speed settings to the user.
- Offline Mode: Progressive web app (PWA) support caching the latest forecast and map tiles for use in areas without cellular coverage.

4. Datasets & Resources

All primary data sources listed below are publicly accessible. These are just recommendations. You may look up and use any other open-source datasets.

4.1 Primary Space Weather Data — NOAA SWPC

Endpoint / Source	Data Provided	Update Cadence	Use For
services.swpc.noaa.gov/products/solar-wind/mag-1-day.json	IMF Bx, By, Bz, Bt vectors from DSCOVR	1-minute	Core nowcast trigger (Bz threshold)
services.swpc.noaa.gov/products/solar-wind/plasma-1-day.json	Solar wind speed, density, temperature	1-minute	Velocity threshold alerts
services.swpc.noaa.gov/json/ovation_aurora_latest.json	360×181 auroral probability grid	~30 minutes	Main map overlay
services.swpc.noaa.gov/products/noaa-planetary-k-index.json	Real-time Kp index	3-hour blocks	Context display and user alerts
services.swpc.noaa.gov/products/3-day-forecast.json	Predicted Kp for next 3 days	2x daily	Advance planning features
services.swpc.noaa.gov/products/alerts.json	Active NOAA geomagnetic alerts & watches	Event-driven	Push notification source

Important API Note

NOAA SWPC updated its JSON schema effective March 31, 2026. Ensure your parsers are compatible with the new formatting. Legacy product endpoints are deprecated. Validate your data ingestion pipeline against the current schema before building downstream logic.

4.2 Meteorological Data

Source	Endpoint / API	Data Available	Notes
Open-Meteo	open-meteo.com	Low / mid / high cloud cover, seeing index, transparency	Free, no API key, astronomy params available
Astrospheric	astrospheric.com (web API)	Astronomy-specific seeing, transparency, cloud layers	Specialized for astrophotography use cases
Meteoblue	https://www.meteoblue.com/weather-api	Astronomical seeing, jet stream, scintillation	More detail on turbulence layers

4.3 Light Pollution & Astronomical Context

Dataset	Source / Format	Use Case
World Atlas of Night Sky Brightness	NOAA VIIRS DNB (GeoTIFF/raster)	Map-based Bortle class estimation per location
Dark Sky Places Database	IDA-maintained GeoJSON	Nearest certified dark sky site routing
Lunar Phase & Illumination	USNO Astronomical Applications API or astronomyapi.com	Moon interference score in Visibility Score
Solar Terminator / Twilight	Calculated via solar zenith angle (SunCalc.js or Ephem Python lib)	Day/night layer and civil/nautical/astronomical twilight bands

5. Evaluation Metrics

Submissions will be judged by a panel on five dimensions. Each dimension is scored 0–20 for a maximum of 100 points.

5.1 Scoring Rubric

Dimension	Weight	What Judges Look For	Full Marks Criteria
Technical Depth & Data Fidelity	20 pts	Quality of real-time data pipeline; correct parsing of OVATION and solar wind JSON; handling of failover, schema changes, and data gaps.	Live pipeline with DSCOVR→ACE failover, correct Bz threshold logic, and documented handling of stale/missing data.
Visualization Quality	20 pts	Accuracy and usability of the auroral probability map. Correct polar projection, dynamic OVATION overlay, day/night terminator, and viewing geometry radius.	OVATION grid rendered as smoothly contoured probability overlay on a globe/map, updating in near-real-time with terminator and viewer radius layers.
Visibility Score Algorithm	20 pts	Soundness and transparency of the composite scoring model. Does it meaningfully integrate cloud cover, darkness, and aurora probability? Is the weighting justified?	Score computes correctly across a range of lat/lon inputs; weighting is documented and defensible; score updates when any input component changes.
UX & Field Usability	20 pts	Dark mode, intuitive alert thresholds, mobile-responsive layout, and fast load time under limited connectivity.	Dark/night-vision mode, sub-3s initial load, mobile-responsive, haptic/audio alerts, and graceful degradation without internet.
Innovation & Stretch Goals	20 pts	Ambition beyond the baseline. GPS Routing, substorm warnings, community sighting layer, photography settings advisor, or offline PWA support.	At least two stretch features demonstrably functional. Novel approach to any part of the problem that meaningfully advances over existing tools.